



GRID FORMING: DROOP CONTROL

ALEJANDRO GIL RESTREPO
ALEXANDER SALAMANCA GALINDEZ
KASPER MICHIELS
KRISTINA GOČANIN
STEWART ROJEC
YORICK VAN DE PAAR

ÉCOLE CENTRALE DE NANTES
DYNAMIC COMPONENTS OF POWER SYSTEMS
MSc. DYNAMICS OF RENEWABLES-BASED POWER SYSTEMS



Tuteur école :
VINU THOMAS
ASSOCIATE PROFESSOR
vinu.thomas@ec-nantes.fr

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Abstract

This work develops a model of a grid forming converter with constant droop control. The first chapter of the work is intended to give a brief overview of the modern power system's challenges while developing some solutions and where grid forming inverters are among them (GFMI from now).

The second chapter of the work describes in detail the structure of the GFMI, including the measuring blocks, primary controller, voltage control loop, current control loop, and the virtual impedance.

The third part of the project outlines the criteria and estimation of the PID controllers used in the voltage and current loops. Time response plots are included as justification for these selections.

The fourth chapter shows the simulation of disturbances to the inverter operation to validate a proper behavior. These include variations in the grid frequency, voltage, and standalone operation disturbances.

Finally, the fifth chapter presents the conclusions that were identified through analysis of the results and the design process.

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1 Introduction

Almost 40% of global greenhouse gas emissions find their origin in energy-intensive industries. These, among others, include the production of electricity and heat [1]. Fossil fuel combustion, which is still widely used for electricity production, is the main source of these greenhouse gas emissions. However, the Paris Agreements signed in 2016 obligated 195 nations to keep the increase in the world's average temperature well below 2 °C [2]. To be able to accomplish this, a fundamental change is needed in our power systems, resulting in rising demand for net zero energy systems. These net zero grids are mainly powered by Inverter Based Resources (IBRs) like solar and wind, instead of fossil fuel powered synchronous machines [3].

The most widely used type of IBRs supply their power through Grid Following Inverters (GFIs). A GFI synchronizes to the grid using a Phase-Locked Loop (PLL) to follow the grid's voltage, phase and frequency. It primarily injects active power into the grid while relying on the grid to maintain voltage and frequency stability. GFI don't support the grid like conventional synchronous machines but follow deviations in frequency and voltage. GFI therefore have a negative impact on grid-strength, also referred to as a loss of inertia [4]. A low strength, or weak, grid can cause network wide voltage and frequency stability issues, with all the associated consequences [5].

A contrary type of IBR device is the Grid Forming Inverter (GFMI). GFMI can operate independently and set the grid's voltage and frequency, mimicking the behavior of a synchronous generator [6]. They provide both active and reactive power to stabilize the grid, even during disturbances. Unlike GFIs, GFMI do not rely on the grid to function and can operate in islanded or weak grid conditions. Research suggests these GFMI will become crucial in maintaining grid stability. Areas spread out over different continents are already experiencing difficulties due to high IBR penetration [7, 8, 9], yet research is still in its early stages [3].

GFMI accomplish this grid-supporting functionality by applying more sophisticated control methods than GFI. A cascade-like control structure consisting of a primary external and secondary internal loop take care of voltage & frequency regulation and current control, respectively, as shown in figure 2.

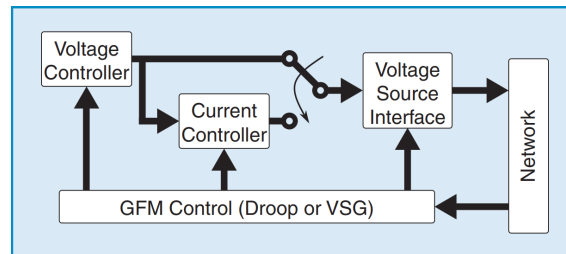


Figure 2: Conceptual GFMI control scheme [3]

The goal of this work is to clarify, build, tune and test a GFMI model using the

Simulink extension of MATLAB. Droop control, stemming from conventional synchronous machine power sharing [5], is chosen as the control method of choice, as it is most widely applied. The basic functioning of GFMI is further elaborated on in section 2, with the measuring blocks, outer control loop, inner control loop and resilience against different grid properties. In section 2, the building and tuning of the model is explained, starting from the chosen inverter and grid parameters and then expanding to how all parameters were tuned. Section 4 goes over the different simulated cases to verify the model, which include weak and strong grid scenarios where the frequency and voltage are deviated. Finally, section 5 concludes this work and goes over the relevant points of discussion.

2 Grid Forming Inverter

This section gives a first introduction to the structure of a GFMI. It introduces each block in the control loop and deduces a specific case that will be further investigated. Figure 3 illustrates a simplified view of the controller. This chapter starts with a generic case setup in which the concept of grid forming is explained and introduces the importance of the $dq0$ -transformation. Secondly, the primary control loop is introduced. Thirdly, the section introduces the inner control loops. This includes the voltage and current control loops, and a method to improve the resilience of the model to changing grid situations. These loops form a chain of control actions of which outputs the terminal voltage level to the PMW generator.

2.1 Measuring Blocks (dq0 transformation)

The dq transformation is a coordinate transformation that maps three-phase quantities (abc) into two quantities with a rotating reference frame (dq) aligned with the system voltage or current. By synchronizing the dq-frame with the grid voltage vector, sinusoidal AC signals are transformed into DC-like quantities in the steady state.

In the stationary abc -frame, the voltages and currents are time-varying sinusoidal signals, making them challenging to control directly. The dq-frame converts these signals into steady-state DC quantities, greatly simplifying control strategies.

The measuring blocks within the GFMI play a critical role in the dq transformation process: coordinate transformation (converts measured abc quantities into the dq-frame for control processing and back to abc for generating inverter output) and filter implementation (reduces high-frequency noise in the measured signals to ensure stability and accuracy of control).

Grid-forming inverters require precise regulation of their output to maintain voltage and frequency stability at the PCC. The dq-frame is essential for achieving this due to decoupled control (so that active and reactive power can be managed in-

dependently, improving the inverter's ability to meet grid demands) and simplified dynamics (DC-like signals in the dq-frame allow for robust and responsive control, enabling smooth operation during transients).

The measured dq quantities are reported to each control loop as additive state feedback terms where appropriate in the control, which will be further detailed in the following subsections.

2.2 Outer Control Loop: Primary Controller

Primary control is a foundational layer in grid-forming inverters. It is responsible for regulating voltage and frequency in real-time to ensure system stability and reliable operation. Unlike GFI, which rely on the grid for synchronization, GFMI establish and maintain voltage and frequency references autonomously. Primary control emulates the behavior of traditional synchronous generators, making it essential for microgrids, weak grids, and systems with high renewable penetration. This capability allows grid-forming inverters to stabilize the grid during disturbances, share power proportionally, and support the overall operation of the electrical system.

The GFMI utilizes a cascaded control architecture that incorporates multiple essential control loops within its operational framework. The key components of this architecture include the inner current control loop, the voltage control loop, the virtual impedance control mechanism, and the primary control loop. These interconnected loops operate collaboratively to regulate the inverter's output power and establish the PCC voltage.

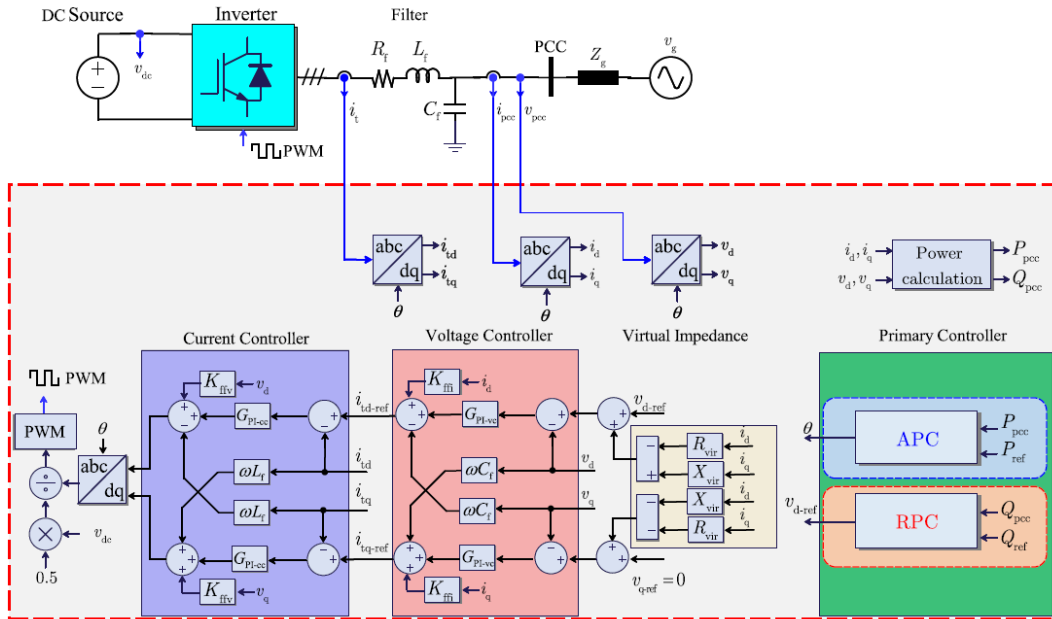


Figure 3: Generic block diagram of the GFMI [10]

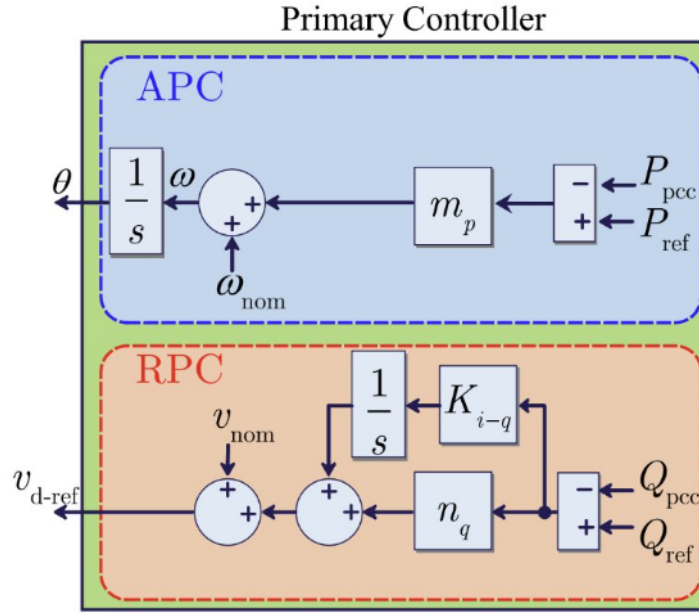


Figure 4: Simple Schematic of a Primary Controller with Conventional Droop [10]

2.2.1 Droop Control: Selected Mechanism of Primary Control

Droop control is the most widely used technique for the primary controller in grid-forming inverters, and is thus the implemented technique in this model [10]. It enables inverters to share power and maintain stability without direct communication. This controller is responsible for providing active and the reactive power sharing. Droop control mimics the natural behavior of synchronous generators, where frequency and voltage drop slightly as active and reactive power demand increase. The inverse proportion between frequency/active power and between voltage/reactive power are the droop coefficients, defined in equations 3 and 4.

$$\omega = \omega_{nom} + m_p(P_{ref} + P_{pcc}) \quad (1)$$

$$V_{d-ref} = V_{nom} + (n_q + \int (K_{i-q} dt))(Q_{ref} - Q_{pcc}) \quad (2)$$

$$m_p = \frac{\Delta\omega}{P_{max}} = \frac{2\pi(f_{max} - f_{min})}{2P_{rated}} \quad (3)$$

$$n_q = \frac{\Delta V}{Q_{max}} = \frac{(V_{max} - V_{min})}{2Q_{rated}} \quad (4)$$

Since the reactive power supplied by the inverter is directly proportional to the voltage between the grid and the inverter [11], a decrease in grid voltage causes an increase in Q supplied by the inverter. The primary controller uses a responding voltage droop to maintain a constant Q supplied by the inverter.

2.2.2 Frequency Droop Control

Frequency droop ensures proper active power sharing among inverters by using the frequency of the grid to track the demand of active power in the same way that synchronous generators do naturally. The frequency deviation is proportional to the error between a nominal active power P_{ref} and the actual output power. As the inverter output power increases as a response to grid demand, $P_{ref} - P_{out}$ becomes negative and droops the frequency down by decreasing a voltage angle θ which is immediately integrated. The exact mechanism of the primary controller responding to active power demand from the grid is explored in the next section.

2.2.3 Grid Frequency Variation

The frequency droop constant in the primary controller causes the inverter to droop the frequency in response to a shortage of active power in the grid, mimicking the behavior of synchronous machines. However, the inverter's droop control is based on active power error, so it is necessary to examine more closely how the inverter responds to a decreased grid frequency with a deviation from the active power reference in the primary controller. Consider the case where the inverter is generating its nominal power at nominal frequency. The inverter controls its internal variables in the dq reference frame, which requires an angle θ that is derived from the frequency of the system. This angle θ allows the dq values to be constant, since it captures the rotating reference frame.

When the grid drops frequency as a response to demanding an excess of active power, its angle becomes different from the angle that is produced by the primary controller. This means that the dq reference frame, which uses the controller angle, no longer rotates at the same angular speed as the actual voltages and currents in the system. Therefore, values in dq are no longer constant and become sinusoidal with the same frequency as the new grid frequency, as shown in figures 5 and 6.

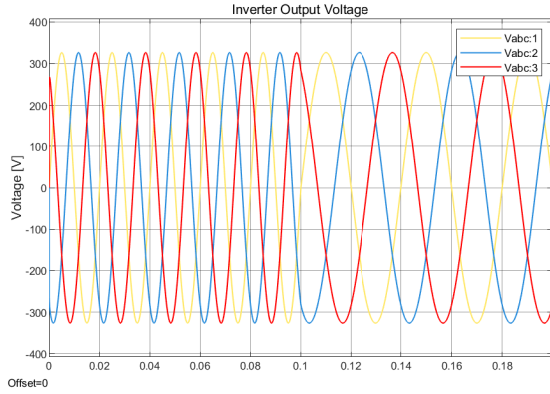


Figure 5: Frequency deviation in abc frame

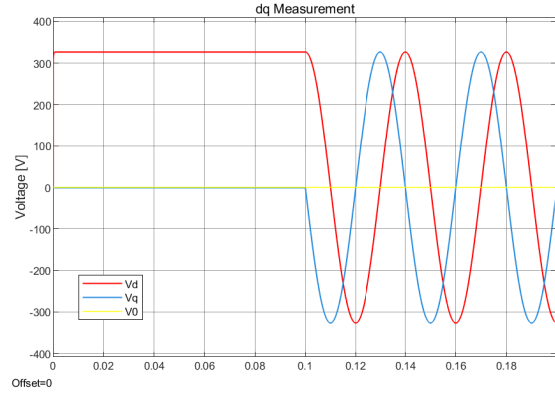


Figure 6: Response to frequency deviation in dq frame

Intuition may indicate that power is also varying with time in the dq frame, thus giving rise to the power variation in the primary controller. However, since current and voltage vary synchronously, power remains constant when no control action is considered.

The voltage and current loops, which are explored in later sections, receive voltage and current feedback from the output. These signals are typically constant in the dq frame but, due to the frequency perturbation, start varying with time. The PID controllers of each loop try to control that deviation, causing an increase of voltages and currents in the dq frame that are given to the PWM generator that feeds the bridges of the inverter as in figure 2. Subsequently, actual currents and voltages of the inverter are modified in the *abc* frame and thereby, actual power is modified. The process is repeated, but in this case, the actual power already changed and the primary controller can identify the deviation in power, which is translated to a deviation in the frequency by the droop constant.

With this frequency deviation, the angle used for the dq transformation in the measuring blocks and the PWM generator are updated, which modifies the modulation wave according to the requirements. The result is the power of the inverter increasing, and the frequency drooping to catch up with the new angle from the grid.

2.3 Inner Control Loops

In the case of angle-droop control, the outer control loop generates the references for the voltage magnitude, V_{dq-ref} , and the voltage angle, θ_{ref} , at the PCC based on the current and voltage measurements at the PCC. These values serve as inputs to the inner control loop analysed in this section. The *q*-component of the reference voltage is often set to zero, so $V_{q-ref} = 0$. As discussed in Section 2.1, all values are converted into the *dq* frame.

The primary objective of the inner control loop is to generate V_{td} and V_{tq} , where

the subscript t denotes the controlled terminal voltages. These voltages, after being converted into the abc frame, are used to calculate the modulation ratio for the PWM control of the inverter.

The core component of the inner control loop is the voltage control loop. Unlike GFIs, which operate as current sources with P and Q control, GFMI function as voltage sources with V and f control. This section implements the voltage control loop for a primary controller operating with angle-based droop control. The primary control loop is designed to regulate the injection of active power P_{PCC} and reactive power Q_{PCC} into the grid to match the desired references P_{ref} and Q_{ref} .

The objective of a GFMI is to dynamically control the magnitude and angle of the voltage at the PCC to achieve synchronism with the grid. By maintaining synchronism, the inverter can also provide support to the grid when necessary. The outputs of the primary control loop must be translated into signals interpretable by the power converter. This is achieved through interconnected control loops. Various topologies can be employed to implement this controller [10], [12].

2.3.1 Voltage-Mode Control

In the first implementation, the voltage loop directly generates $V_{tdq-ref}$, which, after conversion to the abc -frame, is used to produce the modulating signals for the PWM controller. This happens through the control of the error between the reference voltage produced by the primary control loop and the voltage measured at the PCC. The error is controlled in the $dq0$ -reference frame, producing V_{td-ref} and V_{tq-ref} . A simple PI controller reduces the error to zero. The same PI parameters are implemented for both the d and q domain.

This topology does not provide output current control for the inverter, which could lead to over current problems during fault situations [13], [12].

2.3.2 Current-Mode Control

When current limitation is essential, the voltage control loop can be cascaded with a current control loop. In this configuration, the current control loop generates V_{td-ref} and V_{tq-ref} , which are then supplied to the PWM modulation step. Within this work this setup of cascaded control loops is analyzed and implemented.

Figure 7 illustrates the current-mode control topology. In this approach, the voltage control loop is followed by a current control loop, resembling the current control topology of a GFI. The primary control supplies V_{d-ref} , $V_{dq-ref} = 0$, and the voltage angle θ_{ref} to the voltage controller. Using a PI controller, the voltage controller outputs I_{td-ref} and I_{tq-ref} . These values can be limited through overcurrent protection measurements. However, in this work, these measurements are abstracted, and $I_{tdq-ref}$ is directly fed into the current control loop.

Within the current control loop, another PI controller regulates the error between $I_{td\text{-ref}}$ and the measured I_{tdq} at the inverter terminals. The result is then sent to the PWM generation block. Before analysing the transfer functions of the voltage and current control loops, the decoupling of the d and q components is examined to clarify the feed-forward paths.

Decoupling in the dq Reference Frame In the Current Mode Control, the equations in the dq -reference frame are coupled when looking at the AC side. These equations are dependent on the filter impedance.

$$L_f \frac{dI_{td}}{dt} = L_f \omega_{ref} I_{tq} + V_{td} - V_{sd}, \quad (5)$$

$$L_f \frac{dI_{tq}}{dt} = -L_f \omega_{ref} I_{td} + V_{tq} - V_{sq}, \quad (6)$$

$$C_f \frac{dV_{cd}}{dt} = C_f \omega_{ref} V_{cq} + I_{td} - I_{gq}, \quad (7)$$

$$C_f \frac{dV_{cq}}{dt} = -C_f \omega_{ref} V_{cd} + I_{tq} - I_{gd}. \quad (8)$$

Here, L_f and C_f represent the filter inductance and capacitance, respectively. V_{cdq} and I_{gdq} are the measured PCC voltage and current, which are part of a feedforward path that adjusts these values proportionally. A secondary, cross-linked feedforward path facilitates the decoupling of the d - and q -components.

The decoupling for the current control loop is presented below. Switching losses and transmission resistance are accounted for to derive a general expression. A similar approach decouples the d - and q -components in the voltage control loop. ω_0 represents the reference grid angular velocity.

$$L \frac{di_d}{dt} = L \omega_0 i_q - (R + r_{on}) i_d + V_{td} - V_{cd}, \quad (9)$$

$$L \frac{di_q}{dt} = -L \omega_0 i_d - (R + r_{on}) i_q + V_{tq} - V_{cq}. \quad (10)$$

At the terminal of the inverter, the voltage rises to the following levels[12]:

$$V_{td}(t) = \frac{V_{DC}}{2} m_d(t), \quad V_{tq}(t) = \frac{V_{DC}}{2} m_q(t). \quad (11)$$

The modulation signals, m_d and m_q , are expressed below. The new variables, u_{vd} and u_{vq} , represent the PI-controlled errors on the current at the inverter terminals. Since the PI controller operates based on the transmission line impedance, these errors are expressed as voltages.

For decoupling purposes, the dq components, m_{dq} , of the modulation signals are used. However, during PWM generation, these signals are converted into the abc domain.

$$m_d = \frac{2}{V_{DC}} (u_d - L\omega_0 i_q + V_{sd}), \quad m_q = \frac{2}{V_{DC}} (u_q + L\omega_0 i_d + V_{sq}). \quad (12)$$

The decoupled first-order linear systems simplify to [12]:

$$L \frac{di_d}{dt} = -(R + r_{on})i_d + u_d, \quad L \frac{di_q}{dt} = -(R + r_{on})i_q + u_q. \quad (13)$$

Neglecting the switching losses and the transmission resistance simplifies the decoupled equations. The first two equations describe the current controller, and the last two describe the voltage controller [13]:

$$\begin{aligned} L_c \frac{dI_{td}}{dt} &= u_{id}, & C_f \frac{dV_{cd}}{dt} &= u_{vd}, \\ L_c \frac{dI_{tq}}{dt} &= u_{iq}, & C_f \frac{dV_{cq}}{dt} &= u_{vq}. \end{aligned} \quad (14)$$

These equations result in simplified transfer functions [13].

The decoupling strategy is now implemented into the controller for both blocks.

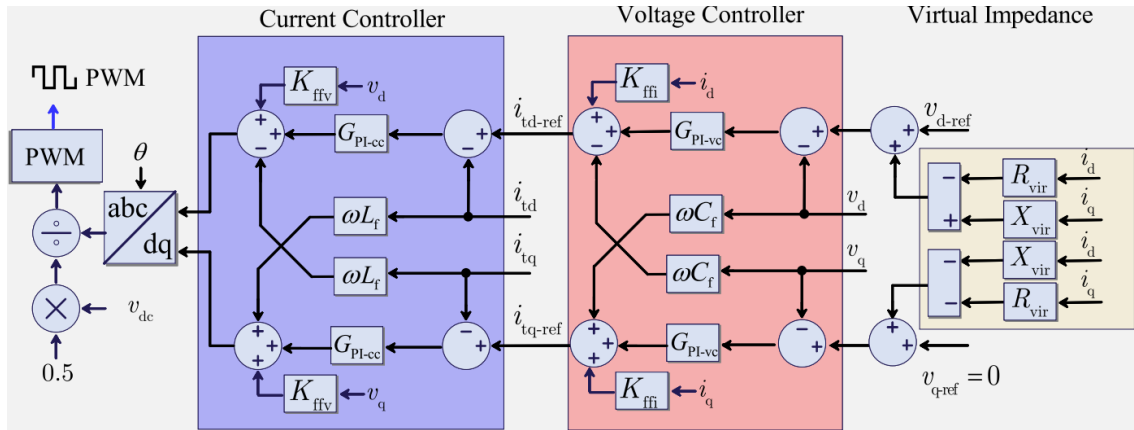


Figure 7: Block Diagram Inner Control Loop [10]

Voltage Control Loop As previously discussed, the voltage control loop uses outputs, either directly or indirectly, from the outer control loops. In this model, the inputs are the calculated V_{d-ref} and the voltage angle θ . While θ is not directly applied within the voltage control loop, all loop inputs, such as the measured current and voltage at the PCC, are transformed into the dq domain using this angle, yielding i_{dq} and v_{dq} .

The voltage angle θ is also used to convert the output of the current controller back to the abc frame, as PWM generation operates using terminal voltage values in the abc frame. The reference value for the q component, $V_{q\text{-ref}}$, is set to zero.

These reference values are compared with the measured v_d and v_q , which have been transformed into the dq frame. The resulting error signals are fed into a PI controller. This controller manages the errors between the measured and reference PCC voltages and is tuned to ensure a closed-loop response that is at least ten times slower than that of the inner current loop. This tuning prevents interference between the two controllers [10]. Both the proportional gain, $K_{p\text{-vc}}$, and the integral gain, $K_{q\text{-vc}}$, must be appropriately configured.

By feeding both v_d and v_q into a proportional term, $\omega \cdot C_f$, where C_f is the filter capacitance, the voltage is converted into a current and used as a feedforward term. This approach decouples the dq components by introducing them into the opposite loop. The resulting feedforward term is added to the output of the PI controller. Figure 7 illustrates this operation, represented as a crosslink at the centre of the controller.

The current components i_d and i_q are also used as feedforward terms to enhance the dynamic response of the inverter [10]. These feedforward paths include a proportional controller in both the d and q domains. The current feedforward coefficient, K_{ffi} , is typically set to 0.9 but can range between 0 and 1.

The feedforward terms are added to the output of the PI controller, along with the subtraction of the decoupling term. This process generates $I_{td\text{-ref}}$ and $I_{tq\text{-ref}}$, which are then forwarded to the cascaded current control loop.

Current Control Loop The subsequent block in the inner control loop is the current controller. This controller receives $I_{td\text{-ref}}$ and $I_{tq\text{-ref}}$ from the voltage controller. When current limitation is required, these signals can be limited using techniques such as current saturation [13].

After applying any necessary limitations, the reference currents are compared to the measured PCC currents. The resulting error is processed by a PI controller which handles the error. The PI controller is characterised by an integral gain, $K_{i\text{-cc}}$, and a proportional gain, $K_{p\text{-cc}}$, both of which must be tuned to achieve a rise time of approximately 1 to 2 ms following a current step [10].

Similar to the voltage controller, the measured PCC currents, I_d and I_q , are fed into a proportional controller, $\omega \cdot L_f$, where L_f is the filter impedance. These currents are converted into voltages and used as feedforward terms to decouple the dq components by feeding them into the opposite loop. The feedforward terms are added to the output of the PI controller. Figure 7 illustrates this operation, represented as a crosslink in the centre of the controller.

Analogously, the current components i_d and i_q are part of proportional feedforward paths that enhance the dynamic response of the inverter in the d and q domains

[10]. The current feedforward coefficient, K_{ffi} , is typically set to 1. The feedforward terms, combined with the subtraction of the decoupling term, are added to the output of the PI controller. This process generates $v_{td\text{-ref}}$ and $v_{tq\text{-ref}}$, which are then fed into the PWM generator.

2.4 Model Resilience Against Variable Grid Properties

In grid-connected systems, the inverter must adapt to varying grid conditions to ensure stable and effective operation. This is critical to maintain the appropriate control of frequency, voltage magnitude, and power flow at the point of common coupling. [14] defines the active power P and reactive power Q exchanged between the inverter and the grid as:

$$P = \frac{3}{R_g^2 + X_g^2} (R_g V_c^2 - R_g V_c V_g \cos \delta + X_g V_c V_g \sin \delta) \quad (15)$$

$$Q = \frac{3}{R_g^2 + X_g^2} (X_g V_c^2 - X_g V_c V_g \cos \delta - R_g V_c V_g \sin \delta) \quad (16)$$

Here, R_g , X_g , V_c , V_g , and δ represent the grid resistance, grid reactance, inverter-side RMS voltage, grid-side RMS voltage, and phase angle difference between the grid and the inverter, respectively. These equations illustrate how the measured P and Q depend on the grid's impedance characteristics, voltage levels, and phase angle difference. To ensure effective inverter operation across varying grid conditions, the coupling between active power and frequency, and between reactive power and voltage, must be managed. Various control techniques exist to achieve this, some examples are adapted droop control, virtual impedance-based control, and H_∞ -based control design [14].

Conventional droop control is optimised for grids with mainly inductive connecting lines. However, deviations in the X/R ratio cause coupling between P and Q , resulting in difficult control. By introducing a transformation matrix that accounts for the X/R ratio, the droop control method is adapted to handle various impedance conditions. This adaptation enables reliable operation in different scenarios and allows the direct regulation of active and reactive currents during faults [15]. The virtual impedance method makes the inverter's apparent output impedance mainly inductive, regardless of the actual line impedance. This is achieved by introducing a virtual inductor, which minimises the coupling between active and reactive power. By reducing the interaction between the two power components, this method simplifies the control process and enhances system stability. The implementation of this approach is straightforward, making it a practical solution for complex grid scenarios [16]. H_∞ -based control offers a systematic design framework for droop controllers in both grid-forming and grid-following inverters. This technique focuses on achieving robust performance across multiple control objectives, such as stability, power

quality, and fault resilience. The method guarantees global small-signal stability in multi-inverter systems, ensuring reliable operation under diverse grid conditions. By employing advanced control theory, H_∞ -based design balances competing demands and achieves optimal performance [17].

The model implemented in this paper will be using the virtual impedance implementation to make the model face fluctuations in the grid impedance. Droop control and virtual impedance are often used together in grid-connected inverter systems to improve performance. Droop control can struggle with power sharing accuracy when line impedances are unbalanced or when nonlinear loads are present. This is because the power-sharing accuracy depends on the line impedance, and there's a tradeoff between accurate power sharing and voltage regulation.

The virtual impedance method helps address these issues by making the inverter's output impedance more inductive. This reduces the coupling between active and reactive power, improving power-sharing accuracy and making the system more stable under varying grid conditions. It also helps manage impedance mismatches without needing large, costly inductive components.

By combining droop control with virtual impedance, the system benefits from both decentralised operation (no communication needed between inverters) and improved power sharing and voltage regulation, even under challenging grid conditions [16].

2.4.1 Virtual Impedance Implementation

By introducing an inductive impedance, the virtual impedance method effectively limits the inverter's current and reduces its sensitivity to grid impedance variations. Figure 7 illustrates the implementation of the virtual impedance within the control system, positioned before the voltage loop. The measured PCC currents, I_d and I_q , are multiplied by the virtual resistance and reactance values of the grid. The resulting d - and q -axis components are combined to produce correction voltages in the d - and q -domains. These correction voltages are added to $V_{d\text{-ref}}$ and $V_{q\text{-ref}}$, generating a corrected reference signal that is fed into the voltage control loop [10].

3 Model Building and Tuning

3.1 The Inverter

Inverters enable conversion of direct current (DC) into alternating current (AC). This functionality is essential for integrating renewable energy sources connected to power converters. This inverter is controlled by PWM which introduces high-frequency switching harmonics into the system due to its pulse operation. There are various types of inverter, each designed for specific applications, with performance and suitability determined by their technical specification. The technical details

of the inverter used for grid-forming are presented in figure 8 (PVU-L0920GR), providing a comprehensive overview of its specifications and capabilities [18].

		PV-PCS			
Type		PVU-L0800GR	PVU-L0840GR	PVU-L0880GR	PVU-L0920GR
Output side (AC)	Rated Power@25°C	800kW	840kW	880kW	920kW
	Rated Power@50°C	730kW	765kW	800kW	840kW
	Rated Voltage	600V +10%, -12%	630V +10%, -12%	660V +10%, -12%	690V +10%, -12%
	Rated Frequency	50Hz / 60Hz (+0.5Hz, -0.7Hz)			
	Rated Power Factor	>0.99			
	Reactive Capability	+/- 421 kVAR	+/- 442 kVAR	+/- 464 kVAR	+/- 485 kVAR
	Rated Current	702 Arms @50 °C			
	Maximum Current	770 Arms @25 °C			
	Maximum Efficiency	98.9% *Tentative			
	CEC Efficiency	98.5% *Tentative			
Input side (DC)	Maximum Voltage	1500 Vdc			
	MPPT Operation Range	875-1300VDC	915-1300VDC	960-1300VDC	1005-1300VDC
Environ. Conditions	Ingress Protection Ratings	IP54 / NEMA3R			
	Installation	Outdoor			
	Ambient Temperature Range	-25° to 50°C			
	Maximum Altitude	>2000 m power derating (Max. 4000m)			
Protective Functions	Input (DC) Side	DC Protection: Fuses Ground Fault, DC Reverse Current, Over Voltage, Over Current			
	Grid (AC) Side	AC Protection: MCCB and Fuse Anti-islanding, Over/Under Voltage, Over/Under Frequency, Over Current			
	Grid Assistance	Reactive/Active Power Control, Power Factor Control, Fault Ride Through (optional)			
Harmonic Distortion of AC Current		≤ 3% THD (at rated power)			

Figure 8: Inverter Used for Grid Forming Inverter “PVU-L0920GR”

3.2 The Grid

3.2.1 X/R Ratio

This parameter determines the behavior of fault currents, including their peak magnitude, waveform asymmetry, and decay rate. The X/R ratio, defined as the ratio of reactance to resistance in a circuit, reflects the influence of the DC component of the fault current, which is assumed to decay exponentially [19].

In power systems, where inductive components, such as transformers, generators, and transmission lines dominate, the X/R ratio has a significant impact on the transient response of the system during fault conditions. A higher X/R ratio results in a slower decay of the DC offset in the fault current, leading to greater asymmetry and increased peak currents. According to [20], a typical X/R value is 6.6 for a system served by a transformer rated at 1000 kVA or 1500 kVA. For this case, the grid-forming has a nominal power of 920 kW, which will be used for the calculations and simulations.

3.2.2 Short Circuit Ratio (SCR)

The SCR is a fundamental metric in power systems used to quantify the strength of the electrical grid at a specific connection point. It is defined as the ratio of the short circuit apparent power (SCMVA) from a three-phase-to-ground (3LG) fault at a given location in the power system to the rated capacity of the inverter-based resource connected to that location. This metric is commonly applied at connection points that involve technologies such as HVDC systems or wind farms, as it helps assess their impact on the system [21].

A low SCR area (referred to as a “weak system”) indicates a high sensitivity of voltage (both in magnitude and phase angle) to changes in active and reactive power injections or consumptions. High SCR (referred to as a “stiff system”) exhibits a low sensitivity and is predominantly unaffected by such changes. Some representative SCR values are shown below, according to [22].

SCR	Classification
SCR > 3	Strong system
2 < SCR < 3	Weak AC system
SCR < 2	Very weak AC system

Table 1: SCR Classification

Nevertheless, SCR is not easily applicable for assessing grid strength when multiple inverter-based resources are connected in close electrical proximity. To address this limitation, several methods have been proposed to estimate the system strength for groups of closely connected inverter-based resources, as follows: [21]

- Weighted Short Circuit Ratio (WSCR).
- Composite Short Circuit Ratio (CSCR).
- Short Circuit Ratio with Interaction Factors (SCRIF).

To account for different systems connected to the grid-forming inverter, Thevenin impedance and SCR are employed to develop the appropriate implementation of the grid-forming inverter using equations 17 - 19. However, it is important to note that using Thevenin impedance is an approximation, as this equivalent cannot capture the varying saturation states of power converters. Therefore, this parameter must be calculated correctly [23].

$$X_{\text{ratio}} = \frac{X_s}{R_s} \quad \text{SCR} = \frac{S_{\text{cc MVA}}}{\text{MW}_{\text{sources}}} = \frac{V_{s_LL}^2}{Z_s \cdot \text{MW}_{\text{sources}}} \quad (17)$$

$$Z_s = \sqrt{R^2 + X^2} \quad R_s = \sqrt{Z_s^2 - X^2} = \frac{Z_s}{\sqrt{1 + X_{\text{ratio}}^2}} \quad (18)$$

$$X_s = X_{\text{ratio}} \cdot R_s \quad (19)$$

Where:

- $S_{\text{cc MVA}}$ = Short circuit 3LG at the connection point.
- $\text{MW}_{\text{sources}}$ = Nominal rated power of the inverter-based resource at the connection point.
- V_{s_LL} = Voltage line-to-line.
- Z_s = Thevenin impedance.
- R_s = Thevenin resistance.
- X_s = Thevenin reactance.

3.3 LCL Filter

As discussed before, inverters introduce high frequency switching harmonics into the system due to their pulse-width modulation (PWM) operation. To mitigate these harmonics, LCL filters are commonly employed between the inverter and the grid. An LCL filter is a passive filter consisting of two series inductors and a parallel capacitor between them. The LCL filter not only improves power quality but also plays a critical role in achieving stable grid synchronization and enhancing system dynamics. Compared to simple L-filters or LC-filters, the LCL filter offers superior harmonic attenuation and with smaller component values. However, the higher-order LCL filter also has a higher capacity to introduce dangerous resonance than an L or LC filter. To combat this, the component values and switching frequency must be chosen with a particular care for the resonant frequency. For additional damping, a resistance connected in series with the capacitance is considered [24].

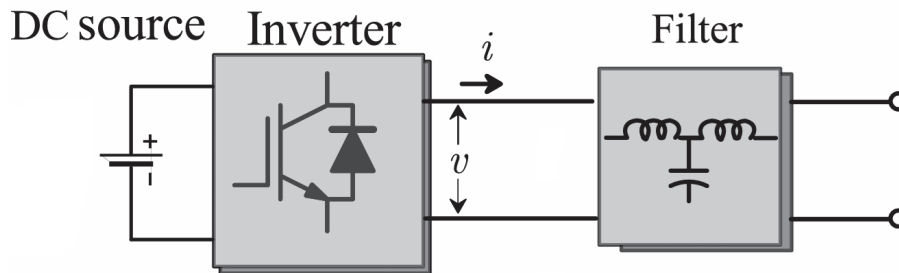


Figure 9: LCL Filter [25]

The procedure for calculating the LCL filter values was based on the previous research [26] and considers a relatively low switching frequency of $f_{sw} = 5000$ Hz. The low switching frequency is justified because the modeled inverter is relatively high power, close to 1 MW, and the switching losses for such an inverter are much higher at more typical frequencies at or above 20 kHz. Additionally, the larger component values needed for a lower frequency can support the higher currents involved.

3.4 Control Loops

3.4.1 Primary Controller

As described previously, the primary controller is based on droop control for both active and reactive power. Droop coefficients can be calculated easily using the inverter rating and the grid's standard parameters.

As mentioned in [27], the maximum frequency deviation of the grid is ± 0.5 Hz. However, most generators are required to remain connected to the grid for frequency deviations of up to ± 1 Hz for at least 30 minutes. This range also applies to isolated areas. Therefore, the droop coefficient is calculated considering the worst-case scenario, as the design's performance is also evaluated in isolated mode. It is given by the equation 3.

In the case of the reactive power loop, low-voltage standards specify that the maximum deviation must not exceed 10% of the rated value. Consequently, the reactive droop constant is defined as in equation 4. Additionally, it is recommended that the reactive power loop include an integrator when connected to the grid. This ensures the ability to follow the reference and supply the required reactive power as determined by the system operator [28]. The gain K_{i-q} associated with this element is set by considering a settling time of 1 second for a step input in the reference. It is important to note that, during standalone operation, the integrator must be disabled to compensate for the reactive power required by the system and the connected load.

3.4.2 Current Control Loop

The current control loop is responsible for comparing the reference and actual current values in the dq frame. According to [25] and [29], the optimal design of the current loop involves using a PI controller, which approximately cancels the pole introduced by the filter. The PI current controller is expressed as in equation 20.

$$k(s) = \frac{k_{pi} \cdot s + k_{ii}}{s} \quad (20)$$

Constants of the PI controller mainly depend on the filter parameters, as shown in equation 21.

$$k_{pi} = \frac{\sum L_f}{\tau_i}; \quad k_{ii} = \frac{R_f}{\tau_i}. \quad (21)$$

τ_i is a design parameter directly related to the performance of the current loop. It should be small for a fast response; however, it must also be large enough such that $\frac{1}{\tau_i}$ remains small compared to the switching frequency of the VSC (in radians per second).

3.4.3 Voltage Control Loop

The voltage control loop is responsible for comparing the reference and actual voltage values in the dq frame. To achieve this, a PID controller is implemented to regulate both voltages and generate the reference current, which is then fed into the current control loop. The PID controller must be designed such that the bandwidth of this loop does not interfere with the current loop's operation (typically 10 times slower). To accomplish this, [25] and [29] propose a method called "Symmetry Optimum", where the proportional gain and the integral time for the PI controller are calculated based on the characteristics of the current loop, and can be easily tuned depending on the desired time response. The PI voltage controller is given by equation 22, whereas the loop gain is given by 23.

$$k(s) = k_{pv} \cdot \frac{s + k_{iv}}{s} \quad (22)$$

$$\ell(s) = \frac{k_{pv}}{\tau_i \cdot C_f} \cdot \left(\frac{s + k_{iv}}{s + \tau_i^{-1}} \right) \cdot \frac{1}{s^2} \quad (23)$$

At low frequencies, the angle is close to 180° . If $k_{iv} < \tau_i^{-1}$, the angle of the closed-loop function initially increases, reaching a maximum δ_m at a certain frequency ω_m , and then decreases asymptotically back to 180° . The phase margin is given by equation 24.

$$\delta_m = \sin^{-1} \left(\frac{1 - \tau_i \cdot k_{iv}}{1 + \tau_i \cdot k_{iv}} \right) \quad (24)$$

$$\omega_m = \sqrt{k_{iv} \cdot \tau_i^{-1}} \quad (25)$$

If $\omega_c = \omega_m$, then δ_m becomes the phase margin. The compensator gain k must satisfy $|\ell(j\omega_m)| = 1$, which leads to the expression 26 for the gain k_{pv} of the PI controller.

$$k_{pv} = C_f \cdot \omega_c \quad (26)$$

As suggested in [29], the phase margin should be between 30° and 75° , and it should be tuned according to the desired response of the grid-forming system.

3.5 Additional Remedial Actions to Improve the Performance of the Grid Forming

Generally, the grid-forming system exhibits good performance when connected to strong grids (with $SCR > 3$ and high X/R ratios). However, it does not perform well when connected to weaker grids. There are two remedial measures to improve performance: introducing a new virtual impedance to make the output impedance much more inductive and modifying the feedforward gains (which range from 0 to 1). The following sections explain how these remedial measures are implemented and the assumptions considered for each.

3.5.1 Virtual Impedance

For the grid shown in figure 10, the active and reactive power flowing from L to C can be expressed in terms of the voltage at both nodes and the impedance between them, as shown in equations 27 and 28.

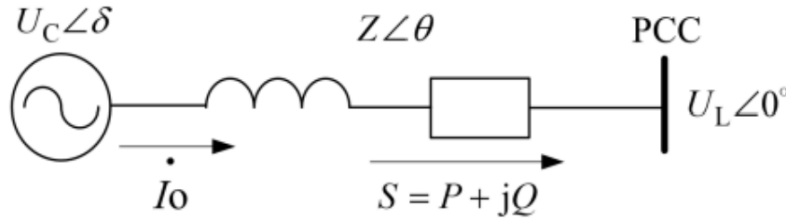


Figure 10: Equivalent Circuit Diagram of Converter Output [30]

$$P_{\text{active}} = \frac{U_L \cdot [R \cdot (U_C \cdot \cos \delta - U_L) + U_C \cdot X \cdot \sin \delta]}{R^2 + X^2} \quad (27)$$

$$Q_{\text{reactive}} = \frac{U_L \cdot [X \cdot (U_C \cdot \cos \delta - U_L) + U_C \cdot R \cdot \sin \delta]}{R^2 + X^2} \quad (28)$$

From equations 27 and 28, it can be seen that there is a significant dependence on the parameters of the grid, R and X , creating a coupling relationship between them. If the resistance is significant compared to the reactance, it is not possible to decouple any term. In general, at lower X/R ratios, the grid's resistance becomes comparable to the grid's reactance, and therefore, active and reactive power cannot be decoupled. If the ratio X/R is high ($X \gg R$), resistance can be neglected, and equations 27 and 28 become equation 29 and 30, respectively.

$$P_{\text{active}} \approx \frac{U_L \cdot U_C}{X} \cdot \sin \delta \quad (29)$$

$$Q_{\text{reactive}} \approx \frac{U_L}{R^2 + X^2} \cdot (U_C \cdot \cos \delta - U_L) \quad (30)$$

According to [30], with these decoupled terms, active power mainly depends on the phase angle δ of the output voltage, and reactive power mainly depends on the output voltage. Moreover, one of the main conditions for grid-forming with droop control to work appropriately is that there should be a direct relationship between active power/frequency and between reactive power/voltage.

As a solution to this problem, virtual impedance has been proposed. Some studies propose only resistances, while others propose resistances and inductances. As the name suggests, a virtual reactance is reflected on the power system, causing a virtual voltage drop, as shown in the following figure.

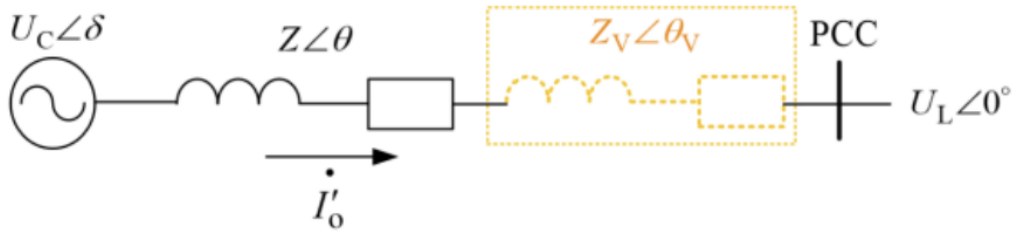


Figure 11: Equivalent Schematic Diagram of the System After Introducing Virtual impedance [30]

Since the system is controlled in the dq reference frame, according to [30], the compensation provided by the virtual impedance should be made as shown in figure 12. The question then becomes how to determine the resistance and reactance values to be compensated in order to achieve a good response in terms of time response, overshoots, robustness, etc. To achieve this, different calculation methods have been proposed, ranging from fixed ranges to optimal values using optimization algorithms [31]. The method will be explained in the following sections.

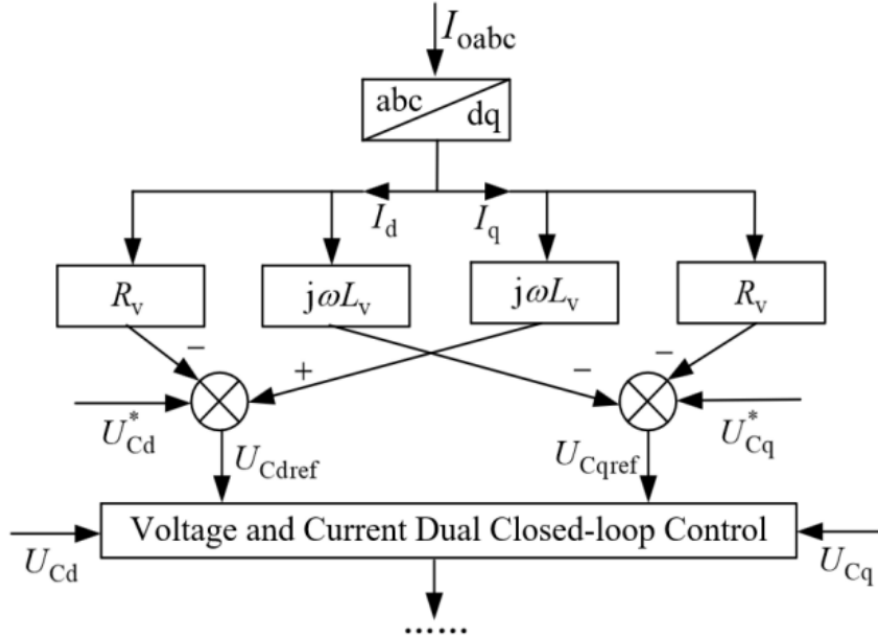


Figure 12: Introduction of Virtual Impedance Schematic Diagram [30]

Another advantage of virtual impedance, in addition to improving the performance of the inverter, is its ability to limit the current contribution when a fault occurs [32]. However, this approach is beyond the scope of this work.

3.5.2 Virtual Resistance

According to [33], the virtual resistance should be designed depending on the grid. By including it, the net resistance of the grid is compensated (see equation 31), and therefore, the grid becomes "stronger" when seen from the inverter. The main idea is to emulate an impedance between the PCC and the inverter output, so that active power is expressed as in equation 32.

$$(-R_{\text{virtual}} + R_{\text{grid}}) \approx 0 \quad (31)$$

$$P_{\text{active}} = \frac{E_{\text{virtual}} \cdot E_S \cdot X \cdot \sin \delta_{\text{virtual}}}{(-R_{\text{virtual}} + R_{\text{grid}})^2 + X^2} \quad (32)$$

Equation 32 suggests that, as long as the resistive effect is avoided, the desired behavior can be achieved. In practice, as [33] states, virtual resistance does not need to offset the line/grid resistance completely, since voltage quality can be improved with smaller virtual impedance.

3.5.3 Virtual Reactance

For the virtual reactance, there is no specific recommendation, and trial and error should be employed to achieve a proper response. However, according to [34], typical values range from 0.03 pu to 0.35 pu. In [35], it is stated that virtual resistance and virtual reactance must meet a specific condition, as shown in the equations 33 and 34.

$$R_{\text{virtual}} = k \cdot R_{\text{grid}} \quad (33)$$

$$(-R_{\text{virtual}} + R_{\text{grid}}) + \omega_0 \cdot (L_{\text{virtual}} + L_{\text{grid}}) > 0 \quad (34)$$

For simplicity, a value within the typical range is chosen and enhanced by trial and error, prioritizing time response and robustness in both weak and strong grids.

3.5.4 Feedforward Gains

As concluded in [36], attenuation of feedforward terms is a useful remedial action to improve the dynamics of the converter. Attenuation of these terms prevents unstable modes by including a constant gain K that varies between 0 and 1.

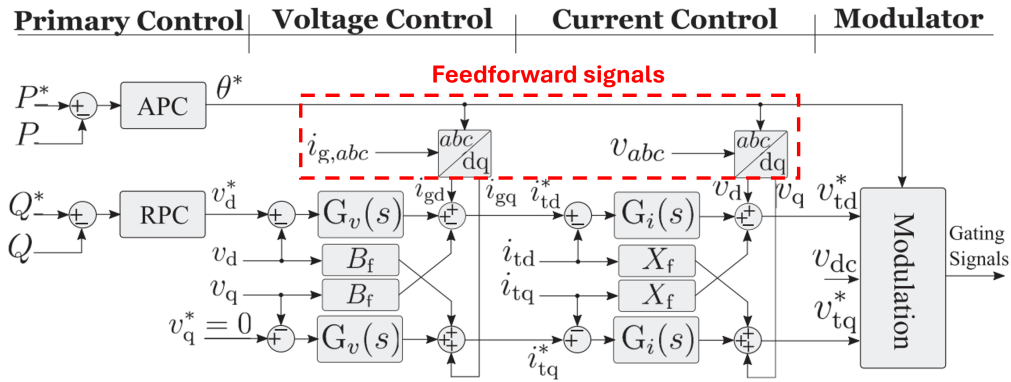


Figure 13: Cascade Control of a GFMI [36]

As suggested by the paper, in this work, the feedforward gains are modified by testing the performance of the grid-forming system with different grids (weak and strong grids).

3.6 Tuning

This section outlines the commissioning process of the grid-forming inverter and highlights the key parameters to consider. Additionally, it presents the different

blocks developed in Simulink and the results of the test simulations

Quantity	Value	Unit	Description
Grid parameters			
v_g	690	V	Grid voltage (L-L, RMS)
f_{base}	50	Hz	Nominal frequency
SCR	1.5	-	Very weak grid
SCR	2.5	-	Weak grid
SCR	7, 10, 30, 60	-	Strong grid
Inverter parameters			
S_{rated}	1.04	MVA	Inverter rated apparent power
P_{rated}	0.92	MW	Inverter rated active power
Q_{rated}	0.485	MVar	Inverter rated reactive power
v_{dc}	1300	V	DC Bus voltage
$LF_{inverter}$	689.43	μH	Filter inductance inverter side
LF_{grid}	98.835	μH	Filter inductance grid side
C_f	61.5	μF	Filter capacitance
$R_{damping}$	0.345	Ω	Damping resistance
R_f	0.01	Ω	Filter resistance

Table 2: Grid and Inverter Parameters

3.6.1 Simulink Model

The following figures provide a detailed overview of the grid-forming inverter model in Simulink and its key components. The figure 14 illustrates the complete model of the grid-forming inverter, showcasing its structure and connections within the system. Figure 15 presents the primary controller and the calculation of active and reactive power. Finally, figure 16 depicts the voltage and current controller loops, emphasizing the control strategies employed to regulate the inverter's output voltage and current, ensuring accurate tracking and system stability.

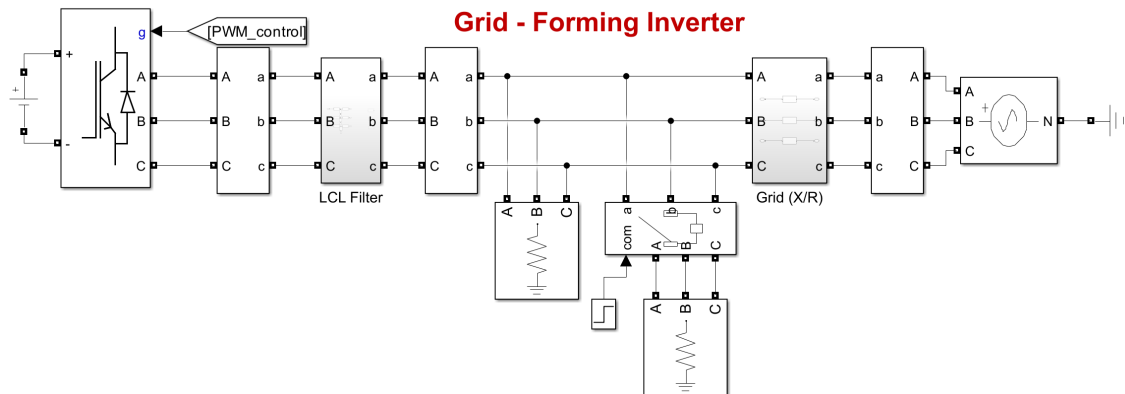


Figure 14: Grid-Forming Inverter

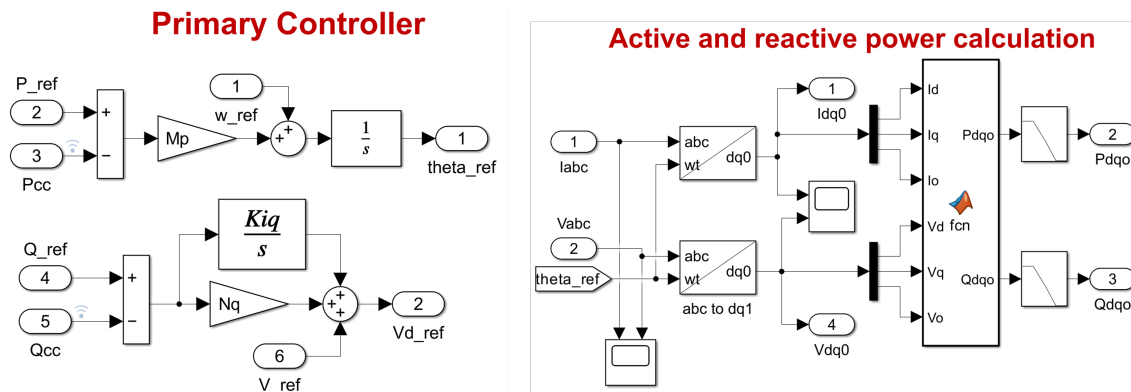


Figure 15: Primary Controller and Active and Reactive Power Calculation

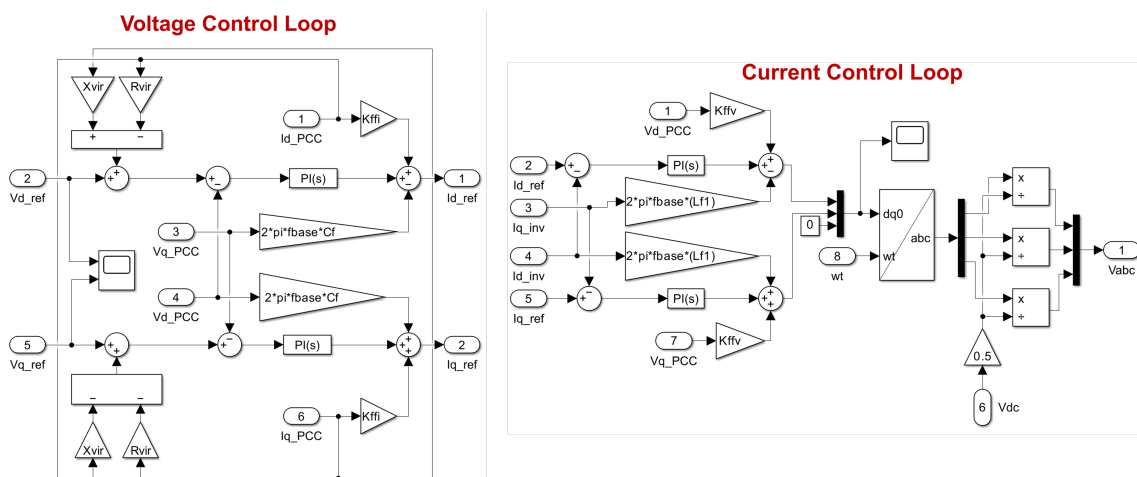


Figure 16: Voltage and Current Control Loop

3.6.2 Test Simulations

Voltage and current are shown in figures 17 and 18.

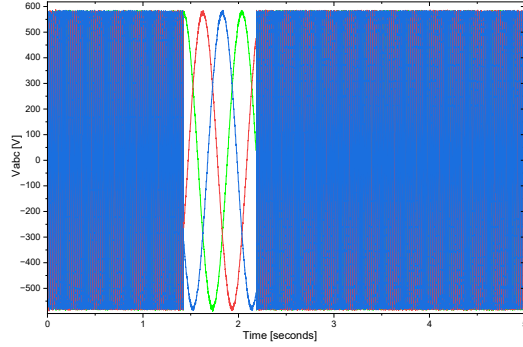


Figure 17: A. Voltage of Phases A, B, and C at the PCC

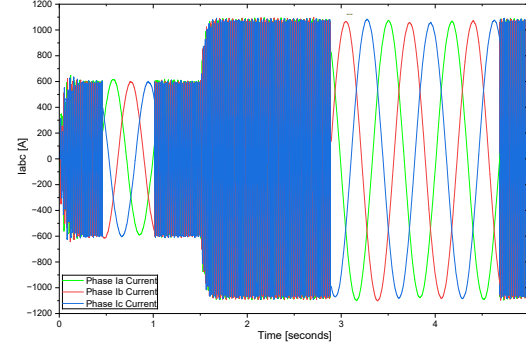


Figure 18: B. Current of Phases A, B, and C at the PCC

Some of the tests are shown below in graphs to give an overview of how the time response varies when controller parameters are changed. The inverter initially generates 500 kW and subsequently, a step of 420 kW is applied to reach its nominal power. The phase margin changes from 30° to 40° in figure 19 and figure 20, the feed-forward gains change from 0.8 to 0.9 in figure 21 and figure 22, and the virtual reactance is varies in figure 23 and figure 24. All of the tests are registered and qualified in table 3.

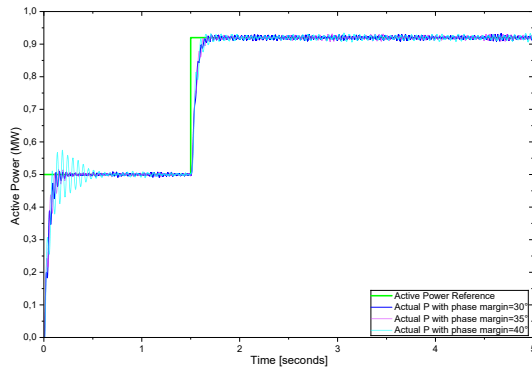


Figure 19: Active Power with Variations of Phase Margin

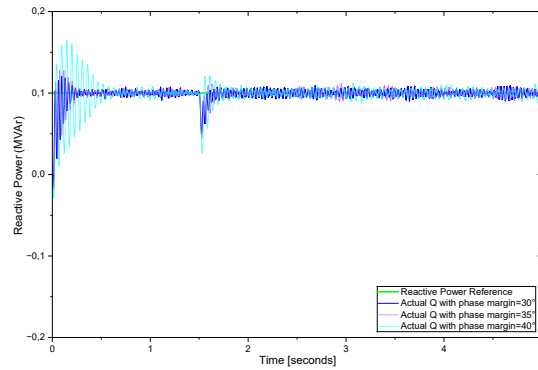


Figure 20: Reactive Power with Variations of Phase Margin

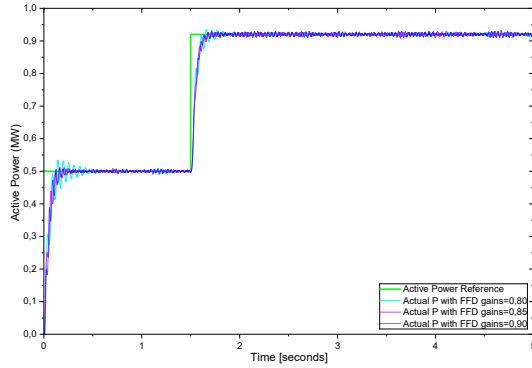


Figure 21: Active Power with Variations of Feedforward Gains

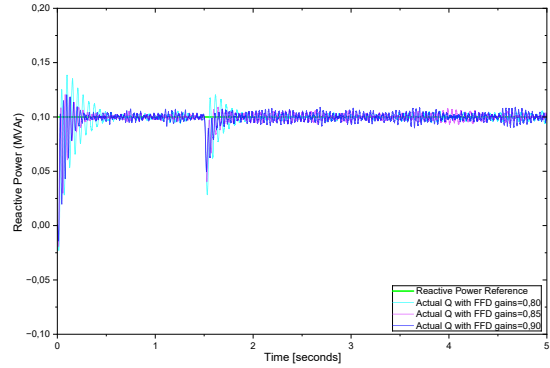


Figure 22: Reactive Power with Variations of Feedforward Gains

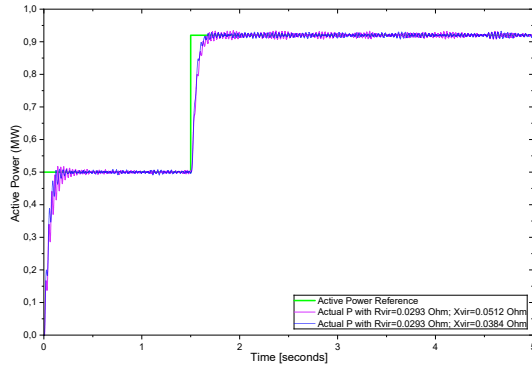


Figure 23: Active Power with Variations of Virtual Reactance

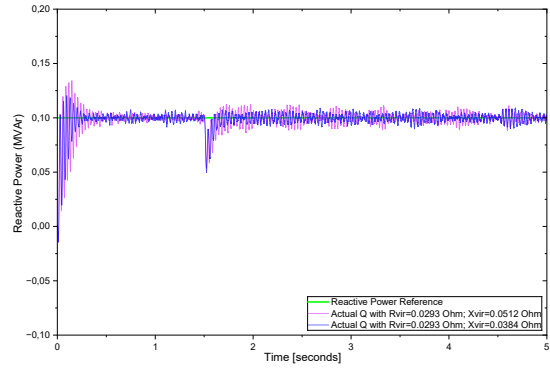


Figure 24: Reactive Power with Variations of Virtual Reactance

Table 3 provides an overview of the system's behavior under varying parameters and conditions. The key parameters examined include the Short Circuit Ratio (SCR), reactance (X_s), resistance (R_s), virtual reactance (X_{vir}), virtual resistance (R_{vir}), inertial time constant (T_i), phase margin (PhM), and voltage and current feed-forward gains (K_{ffv} and K_{ffi}). These parameters were systematically adjusted to analyze their impact on the system's performance for each configuration.

The stabilization time is used as a key performance indicator, measuring how quickly the system reaches a steady state when the active power achieves 0.49 MW, based on a 0.5 MW active power reference. The observations also assess the waveform's stability, ensuring that no overshoot or instability is present.

The study incorporates more than 30 simulation scenarios, providing a comprehensive analysis of the system's response under diverse conditions and parameter variations. This extensive approach demonstrates the robustness of the control strategy while identifying critical factors that influence system performance. The final values and the best system response, determined by time behavior and steady-state performance, are highlighted in green for clarity.

SCR	Xs	Rs	Xvir	Rvir	Ti [ms]	PhM	Kffv	Kffi	Stabilization Time [ms]	Response
2	0.2561	0.0366	0.0845	0.0293	0.2	53	1	0.9	800	Good
7	0.0732	0.0105	0.0845	0.0293	0.2	53	1	0.9	Overshoot++	Overshoot++
10	0.0512	0.0073	0.0845	0.0293	0.2	53	1	0.9	Overshoot++	Overshoot++
2	0.2561	0.0366	0.0512	0.0293	0.2	53	1	0.9	700	Good
7	0.0732	0.0105	0.0512	0.0293	0.2	53	1	0.9	280	Good
10	0.0512	0.0073	0.0512	0.0293	0.2	53	1	0.9	250	Good
2	0.2561	0.0366	0.0384	0.0293	0.2	53	1	0.9	700	Good
7	0.0732	0.0105	0.0384	0.0293	0.2	53	1	0.9	250	Good
10	0.0512	0.0073	0.0384	0.0293	0.2	53	1	0.9	180	Good
2	0.2561	0.0366	0.0384	0.0293	0.2	50	1	0.9	670	Good
7	0.0732	0.0105	0.0384	0.0293	0.2	50	1	0.9	250	Good
10	0.0512	0.0073	0.0384	0.0293	0.2	50	1	0.9	155	Good
2	0.2561	0.0366	0.0384	0.0293	0.2	50	0.9	0.9	670	Good
7	0.0732	0.0105	0.0384	0.0293	0.2	50	0.9	0.9	240	Good
10	0.0512	0.0073	0.0384	0.0293	0.2	50	0.9	0.9	145	Good
30	0.0171	0.0024	0.0384	0.0293	0.2	50	0.9	0.9	Overshoot++	Overshoot++
60	0.0085	0.0012	0.0384	0.0293	0.2	50	0.9	0.9	Overshoot+++	Overshoot+++
2	0.2561	0.0366	0.0384	0.0293	0.2	30	0.9	0.9	700	Good
7	0.0732	0.0105	0.0384	0.0293	0.2	30	0.9	0.9	240	Good
10	0.0512	0.0073	0.0384	0.0293	0.2	30	0.9	0.9	170	Good
30	0.0171	0.0024	0.0384	0.0293	0.2	30	0.9	0.9	90	Good
60	0.0085	0.0012	0.0384	0.0293	0.2	30	0.9	0.9	60	Overshoot-
2	0.2561	0.0366	0.0384	0.0293	0.2	50	0.8	0.8	665	Good
7	0.0732	0.0105	0.0384	0.0293	0.2	50	0.8	0.8	Overshoot	Overshoot
10	0.0512	0.0073	0.0384	0.0293	0.2	50	0.8	0.8	Overshoot++	Overshoot++
2	0.2561	0.0366	0.0384	0.0293	0.1	50	1	0.9	700	Good
7	0.0732	0.0105	0.0384	0.0293	0.1	50	1	0.9	Overshoot	Overshoot
10	0.0512	0.0073	0.0384	0.0293	0.1	50	1	0.9	Overshoot	Overshoot
2	0.2561	0.0366	0.0384	0.0293	0.2	46	1	0.9	650	Good
7	0.0732	0.0105	0.0384	0.0293	0.2	46	1	0.9	250	Good
10	0.0512	0.0073	0.0384	0.0293	0.2	46	1	0.9	180	Good
2	0.2561	0.0366	0.0384	0.0293	0.2	60	1	0.9	650	Good
7	0.0732	0.0105	0.0384	0.0293	0.2	60	1	0.9	Overshoot++	Overshoot++
10	0.0512	0.0073	0.0384	0.0293	0.2	60	1	0.9	Overshoot++	Overshoot++

Table 3: Testing and Performance Simulations ($X/R = 6.6$)

3.6.3 Tuning Results

The table 4 provides a comprehensive overview of the control parameters used for grid-forming control strategies, specifically focusing on inner loop control, droop control, and relevant virtual impedance parameters. These parameters play a crucial role in maintaining stability and optimizing the system's dynamic response.

Quantity	Value	Unit	Description
Control parameters for the inner loops			
$k_p - cc$	3.9413	Ω	Proportional gain of the current controller
$k_{ip} - cc$	50	Ω/s	Integral gain of the current controller
k_{ffv}	0.9	-	Voltage feed-forward
k_{p-vc}	0.1776	S	Proportional gain of the voltage controller
k_{i-vc}	295.9353	S/s	Integral gain of the voltage controller
k_{ffi}	0.9	-	Current feed-forward
R_{vir}	0.0293	Ω	Resistance of virtual impedance
X_{vir}	0.0384	Ω	Inductance of virtual impedance
Droop control			
m_p	3.4148×10^{-6}	rad/W.s	$P - \omega$ Droop coefficient of the APC
n_q	1.1616×10^{-4}	V/Var	$Q - V$ Droop coefficient of the RPCC
k_{i-q}	7×10^{-3}	V/Var.s	Integral gain of the RPC

Table 4: Controller Parameters Grid-Forming

4 Simulation Results

The process for grid forming control is validated by simulating various grid events with the modeled grid in Simulink, and measuring the response of the implemented control loops as well as the response of salient grid and PCC features. The primary control loop begins with a comparison to an active and reactive power reference, so the grid disturbances tested will be a frequency and voltage deviation. Additionally, the model will be tested in an islanded mode with variations in power demand from a local load.

In the grid-connected operation, the operating point from which the grid will deviate is selected to be within the acceptable operating range of a typical inverter used for a DC renewable source such as 8. The grid voltage and base reference voltage of the inverter is 690V phase to phase, or 400V phase to neutral, which is what is most commonly measured. It will normally supply a P_{ref} of 500 kW and a Q_{ref} of 100 kVAR at a base frequency of 50 Hz. Additionally, each deviation will be tested on both a "strong" grid, simulated with an SCR (section 3.2.2) of 10, and a "weak" grid with an SCR of 2.

4.1 Frequency Deviation Response

First, the common grid event of a frequency drop is simulated. When more active power is demanded of the grid than is available from it, the synchronous machines slow down accordingly, thus maintaining conservation of energy and dropping the grid frequency. The grid forming inverter should respond to this frequency drop with an increase in the supply of active power to fulfill its control objective of supporting

the grid.

In the simulation testing this scenario, the frequency of the programmable voltage source representing the grid is decreased from by a step of 0.25 Hz at 0.5s for 0.2s, and the response of the system is measured over 1.3 s to allow transients to settle before and after the disturbance. The weak grid's deviation is simulated at 0.6s for 0.4s due to its longer transient response. The results of the simulation at the PCC are shown in figures 25 and 26.

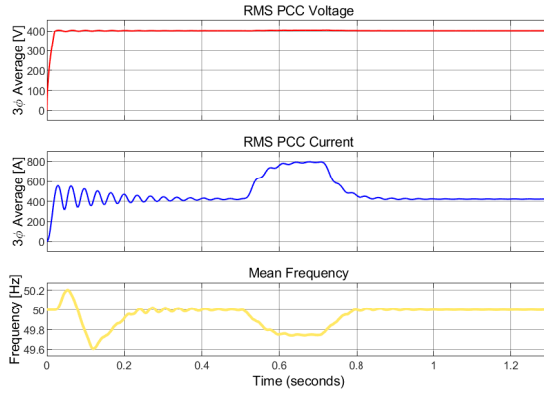


Figure 25: PCC Response to Strong Grid Frequency Deviation

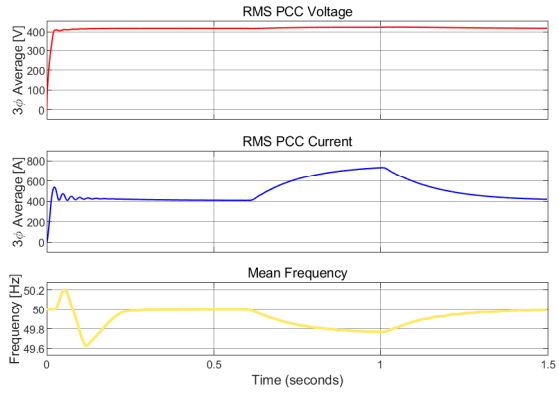


Figure 26: PCC Response to Weak Grid Frequency Deviation

What is first noticeable is the startup transients that take about half of a second to clear before settling into the specified operating point. Since the inverter starts completely cold, there is a large inrush current and voltage rise at startup. Additionally, the power references are controlling the inverter from the beginning, which causes the current oscillations and frequency fluctuation from the droop control. After the startup transients settle, the grid voltage source drops the frequency at 0.5 s, and the inverter frequency responds by drooping the frequency while increasing the current. The weak grid case has a much slower response, since it has a much higher impedance and therefore inductance.

The response at the grid is shown in figures 27 and 28.

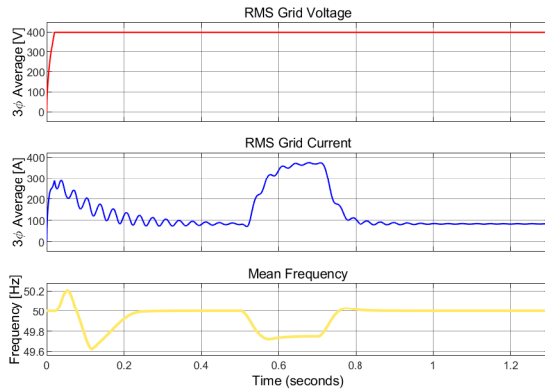


Figure 27: Weak Grid Response to Frequency Deviation

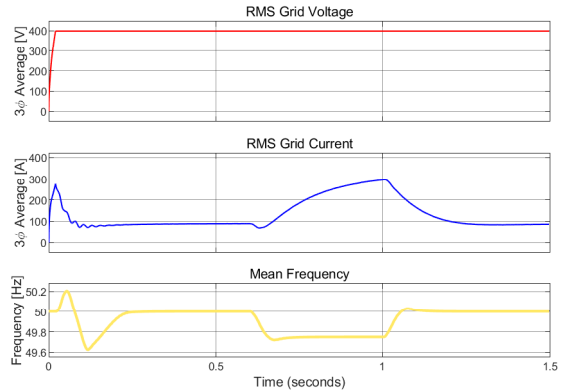


Figure 28: Strong Grid Response to Frequency Deviation

The grid displays a similar startup transient to the PCC, only with the inrush current decreasing to the operating point. The current and the frequency responses are also similar, which is expected as only a small impedance representing a transmission line separate them. However, it is noticeable that the frequency of the grid decreases much quicker than the PCC. The frequency drop is initiated from the source at the grid, so the hesitancy of the PCC frequency to drop indicates that the control scheme provides the inertia that is intended from the grid-forming control objective, all while maintaining a constant voltage. Again, the weak grid case presents analogous behavior at a much longer settling time due to the low SCR.

The next captured behavior is the active and reactive power injected by the inverter at the PCC, before the transmission line impedance to the grid. The response is shown in figures 29 and 30.

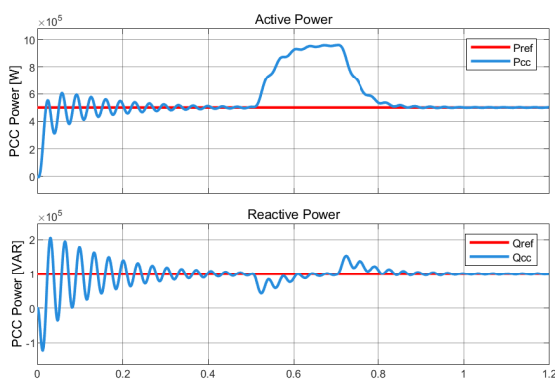


Figure 29: Strong Grid PCC Power Under Frequency Deviation

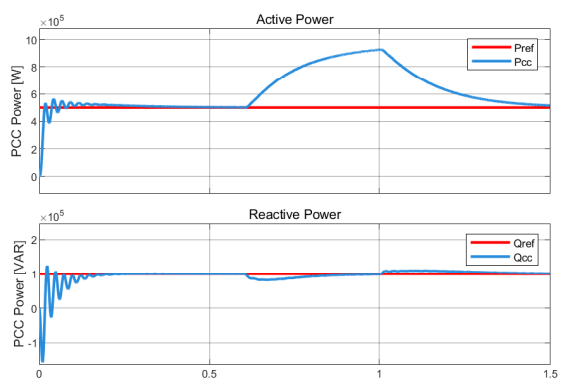


Figure 30: Weak Grid PCC Power Under Frequency Deviation

Similar startup transients are visible due to the combination of the voltage and current responses, but it is clear that the inverter tracks both operating point references

properly in the region before the disturbance. After the disturbance, the inverter responds by injecting more active power, almost 400 kW more. The additional active power is in line with the control objective, since a decrease in frequency from the grid would indicate that more active power is required. Since this grid model is only a voltage source that does not respond to the increased power provided, the inverter reaches a steady state of increased power proportional to the frequency change and the droop constant. The frequency drop represents a 0.5% drop, combined with the realistic droop constants [36] results in the high active power increase. A more realistic grid model would respond to the increase in power with a restoration of the frequency, and the inverter would reach a lower steady state.

The reactive power response shown in figure 29 is also notable. Since reactive power is of course directly related to frequency, the initial decrease after the frequency disturbance is expected, as is the inverse after the disturbance clears. The control output of the primary control loop that regulates Q is the voltage reference, which can be seen in figures 31 and 32.

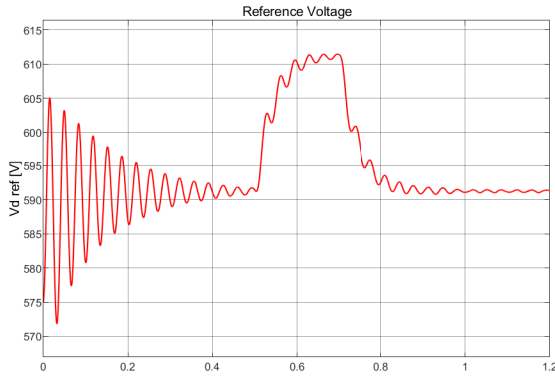


Figure 31: $V_{d_{ref}}$ Under Strong Grid Frequency Deviation

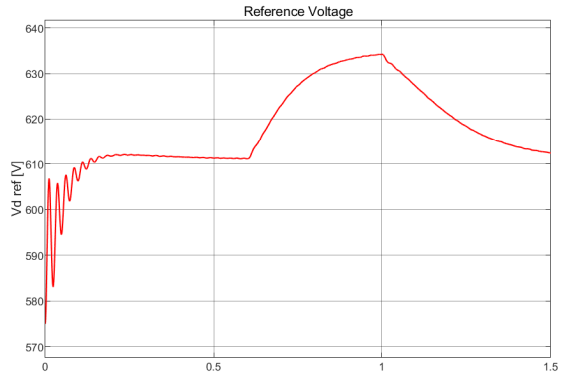


Figure 32: $V_{d_{ref}}$ Under Weak Grid Frequency Deviation

The increase to compensate for the lost Q is expected and proportional to the Q-droop constant.

4.2 Voltage Deviation Response

The second scenario that is simulated is a drop in grid voltage. A drop in voltage is an indicator for a demand in reactive power in the grid. Normally, a synchronous machine would respond to such a demand by overexciting the rotor-current and thereby outputting a higher voltage at the PCC, supplying the grid with reactive power.

In the simulation a voltage step of -0.15 pu is applied at the grid at 0.5s for 0.2s. This value is slightly higher than the permitted range shown in figure 8, chosen to better show the response behaviors while staying close to normal operating values.

Expected behavior is that the inverter responds to the drop in voltage by supplying the demand in reactive power. Again, the weak grid simulation has the same step applied over a longer time period due to allow for complete observation of its slower response. Figure 33 shows the voltage, current and frequency response at the PCC.

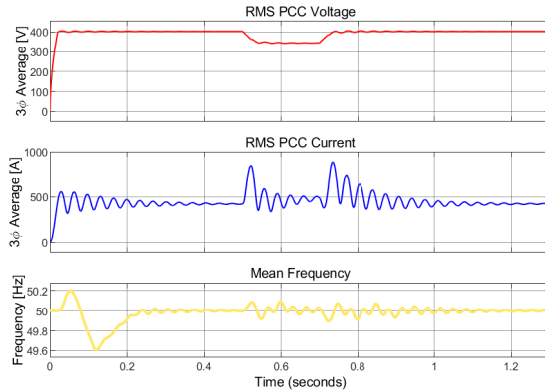


Figure 33: PCC Response to Strong Grid Voltage Deviation

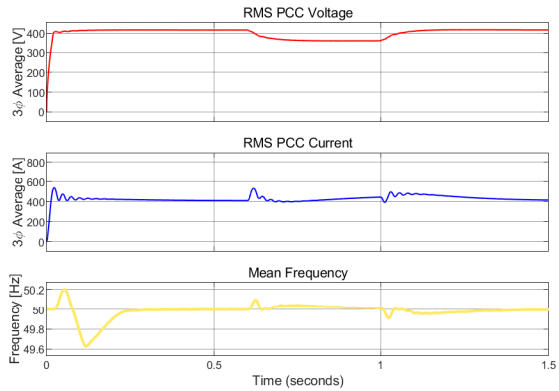


Figure 34: PCC Response to Weak Grid Voltage Deviation

The same initial transient from startup is observed as in the previous scenario. After this transient settles down the voltage deviation is applied. The voltage at the PCC is seen to drop together with the voltage drop simulated at the grid. It drops notably slower than the grid voltage as shown in figure 36, indicating the same inertial behavior as with the decrease in frequency. Due to the drop in voltage, the current increases to supply the power demand accordingly. The frequency shows oscillations due to the transient behavior, which settle around the nominal value.

Figure 36 shows the grid behavior in response to the voltage drop. As expected, the voltage drops as programmed in the source. A similar response in current is observed at the grid, only of lower magnitude. This can be attributed to the impedance behavior of the line between the PCC and the grid. The frequency shows minor distortions, but no lengthy oscillations as seen at the PCC, as the grid maintains its nominal value due to its strong and stable operation.

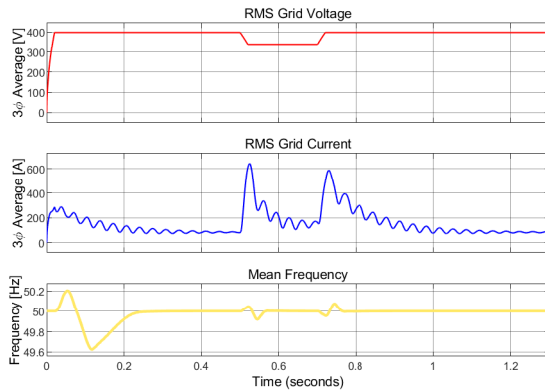


Figure 35: Strong Grid Response to Voltage Deviation

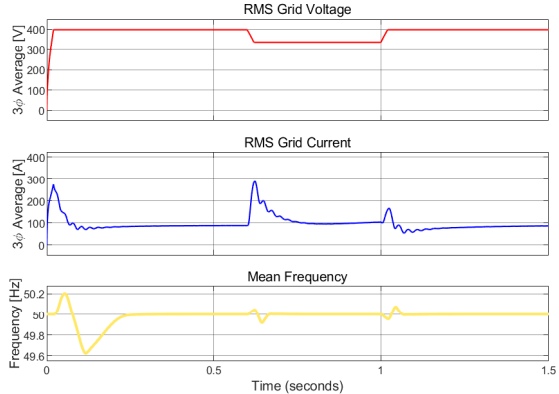


Figure 36: Weak Grid Response to Voltage Deviation

In figures 37 and 38 the active and reactive power are shown in response to the voltage step. As expected, the inverter responds to the drop in voltage by increasing its reactive power output. In figures 37 and 38 the response in voltage reference in the d-frame is shown. When the voltage drop is initiated the reference is seen to drop. At first glance this seems to be undesired, as intuitively the inverter should supply reactive power for the duration of the voltage decrease. However, the Q_{ref} is determined by a higher control loop, and this primary controller is designed to maintain Q by drooping voltage through disturbances.

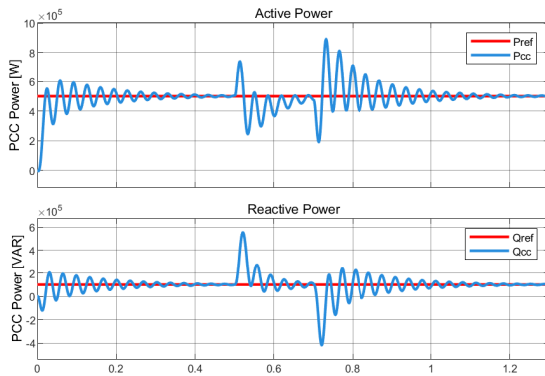


Figure 37: Strong Grid PCC Power Under Voltage Deviation

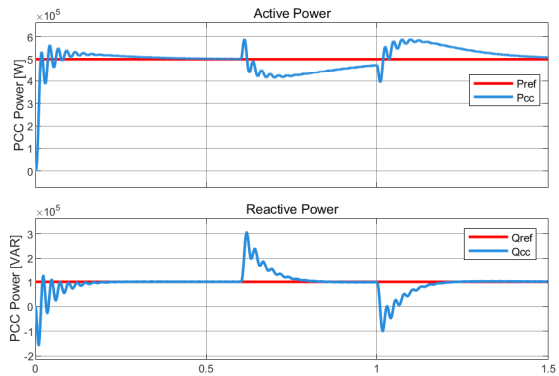


Figure 38: Weak Grid PCC Power Under Voltage Deviation

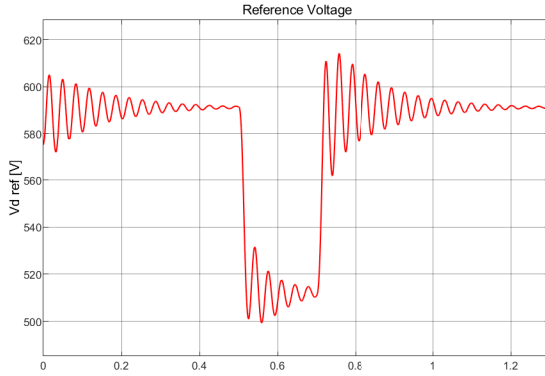


Figure 39: $V_{d_{ref}}$ Under Strong Grid Voltage Deviation

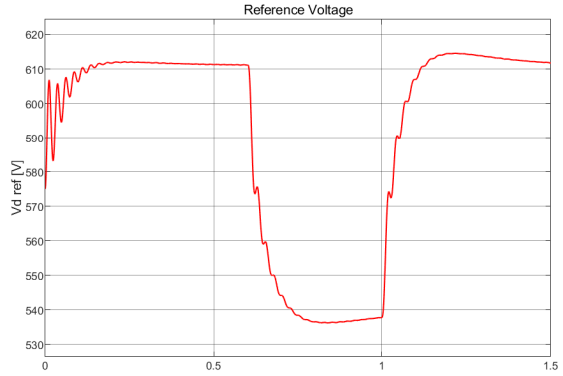


Figure 40: $V_{d_{ref}}$ Under Weak Grid Voltage Deviation

From figure 15 it can be seen that the Q_{cc} is subtracted from Q_{ref} before it is fed to the secondary control loop. Hence, when the reactive power output spikes in response to the increased voltage drop over reactive transmission line, the negative control output reduces the reference voltage $V_{d_{ref}}$ accordingly. In order for the inverter to keep supplying reactive power during a voltage drop the reference signal should be adjusted by over-coupling control of the inverter outside of the scope of this model. The active power shows an increase resulting in oscillations due to the observed peak in current.

4.3 Standalone Mode

The final examined behavior of the grid forming inverter model is under isolated conditions with local loads. In these simulations, the transmission line and grid voltage source are removed and replaced with a constant impedance load, which also defines the P_{ref} and Q_{ref} for the inverter's standalone operating point. The grid deviations are replaced by an additional constant impedance load added in parallel to the initial one, representing a disturbance more realistic to a microgrid.

4.3.1 Resistive Load

The first situation modeled is a purely resistive nominal load that consumes 500 kW of active power. At 0.5 s, an additional resistive load of 200 kW is added in parallel. The results at the PCC are shown in 41.

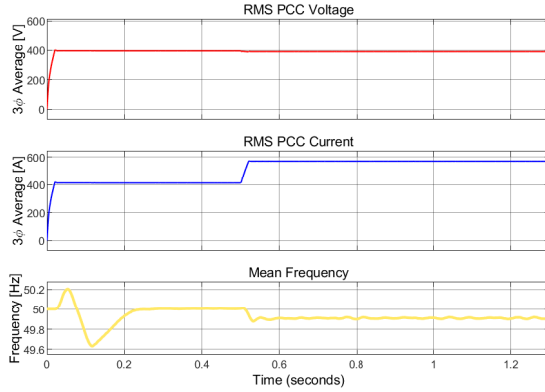


Figure 41: PCC Response to Standalone Resistive Load Step

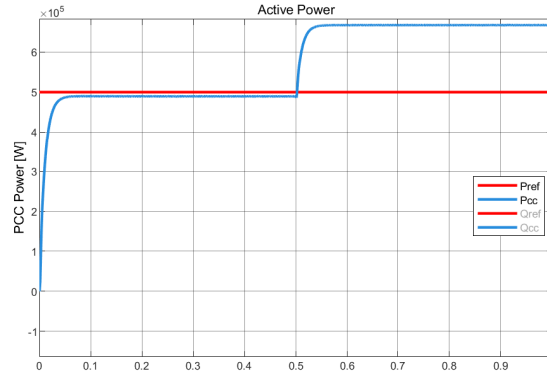


Figure 42: Power Response to Standalone Resistive Load Step

The same startup transients in the frequency are visible, indicating that they are inherent to the control loop, although it does stabilize about twice as fast in standalone mode. The voltage and current also reach a steady state quickly corresponding to the operating point. After the load increase, the current increases slightly to match the increased power demand as expected, while the voltage remains regulated. The frequency drops in accordance with the droop control. The grid forming inverter is correctly mimicking the inertia of a synchronous generator, which would also experience a frequency drop in the event of such a load increase. Restoring the frequency under the new load conditions would require increasing the P_{ref} , which would be dictated by a higher control loop outside the scope of this simulation.

The power output of the inverter is shown in figure 42. The power output rises to the reference at startup, and then rises by the amount of the increased load at 0.5 s as expected. It is noticeable that the actual power output is slightly below that of the reference, which is attributable to the change in dynamics of power flow in the system. The droop controller for the active power controls the angle of the voltage from the inverter, but the power delivered by the inverter no longer depends on the angle in a standalone mode. The power supplied is therefore distorted by terms in the control loop relevant to the grid-connected case like the virtual impedance and P/Q decoupling terms. The virtual impedance no longer serves the same function of increasing the X/R ratio in this purely resistive scenario, since the ratio is close to zero.

5 Conclusions & Discussion

This study presents a thorough analysis of the GFMI operating principles and presents a working model of such a system, simulates its performance through over 30 scenarios, assessing the impact of diverse parameters and operating conditions. The results highlight the robustness of the proposed control strategy, effectively ad-

addressing the dynamic behavior and steady-state performance of the system. There are several key takeaways from this study, first of which is the virtual impedance.

The most significant advantage of virtual impedance, among others, is that it makes the system more stable by modifying the X/R ratio of the system without introducing additional losses. Nevertheless, it is important to consider voltage quality when selecting the values to be compensated. From the evaluations conducted in this work, it can be noted that larger virtual inductance leads to longer settling times, potentially making the system unstable. This conclusion becomes particularly relevant when the GFMI operates at low SCRs, where the settling time is considerably longer.

Additionally, attenuation of feedforward terms is a useful remedial action for improving the dynamics of the converter. These terms mitigate unstable modes by incorporating a constant gain K , which stabilizes the control system. By combining these terms with the phase margin and the current loop time constant, it becomes possible to manage disturbances effectively. Moreover, enhancing response accuracy contributes to improved overall performance and robustness of the converter under varying operating conditions, as shown in table 3.

Several considerations should also be made for future research into modeling and controlling GFIMs. Key findings indicate that overshoot can be mitigated by reducing the phase margin or by tuning the LCL filter design. Improvements to the simulation are also possible. The grid model considered for this work is an infinite bus; however, it is important to point out that it does not capture any dynamic response to injected power from the inverter. Moreover, the grid's characteristics change depending on the connection point, so it is advisable to design and evaluate considering a proper equivalent that captures all these effects at the connection point.

Finally, the results of the standalone simulation indicated that the proposed control scheme may be less suitable for isolated local loads, and is a good topic for future research. The results section presented purely resistive local load, as would be a common for an islanded case, but local reactive power demand produced unpredictable results from the model, including poor voltage tracking and instability. Other research [37] includes an additional integrator to improve voltage tracking in conditions of varying reactive power.

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